

## **MEANING OF LOGISTICS**

Logistics is the process of planning, implementing and controlling the flow of goods and services from the place of production to the place of consumption to satisfy customers. It ensures that the right product reaches the right place, at the right time, in the right quantity and condition. It includes transportation, warehousing, packaging, material handling, inventory management and information management.

## **IMPORTANCE OF LOGISTICS**

### **1. Backbone of Trade and Business**

Trade and business require the movement of goods from producers to markets and customers. Without logistics, buying and selling of goods cannot take place smoothly. Hence, logistics is the backbone of trade and business.

### **2. Leads to Customer Satisfaction**

Logistics helps deliver goods of the right quality and quantity at the right time, place and condition. This helps meet customer expectations and increases customer satisfaction.

### **3. Cost Reduction and Profit Maximization**

Efficient logistics helps reduce costs through better material handling, economical transportation and proper warehousing. Lower costs ultimately help in increasing profits.

### **4. Competitive Advantage**

A company that provides fast and reliable delivery can gain an advantage over its competitors. Logistics helps improve customer service and reduce errors in order processing.

### **5. Effective Communication System**

Logistics helps maintain continuous communication between suppliers, the business and customers. It also helps the business respond quickly to customer requirements.

### **6. Supports Operations and Marketing**

Logistics supports operations by ensuring the availability of raw materials for uninterrupted production. It supports marketing by ensuring that finished products are properly packed and delivered to customers at the right time and place.

## **BAD LOGISTICS**

Bad logistics means logistical activities are not performed properly according to the required standards. It occurs when actual logistical performance is different from the expected performance.

### **CAUSES OF BAD LOGISTICS**

#### **1. Inadequate Infrastructure:**

Lack of proper roads, warehouses, transportation facilities, vehicles and other infrastructure affects the smooth movement and storage of goods.

#### **2. Taxation:**

High or complicated taxes can increase the cost of logistics. Different taxes and procedures may also cause delays in the movement of goods.

#### **3. Inadequate Information:**

Lack of correct and timely information can result in wrong decisions, poor coordination and delays in delivery.

#### **4. Poor Management Decisions:**

Improper planning, wrong decisions and lack of coordination by management can make logistics activities inefficient.

### **EFFECTS OF BAD LOGISTICS**

#### **1. Costly Products:**

Bad logistics increases transportation, warehousing, handling and other costs. These increased costs may make the final products expensive.

#### **2. Damaged Products:**

Improper packaging, handling and transportation can cause damage to goods before they reach the customer.

#### **3. Excessive Movement Costs:**

Poor planning may lead to unnecessary movement of goods, repeated transportation and higher transportation costs.

#### **4. Shortages:**

Delays and poor inventory management can result in insufficient supply of goods in the market.

#### **5. Black Marketing:**

Shortage of goods caused by poor logistics may encourage sellers to sell products at higher prices through black marketing.

#### **6. Exploitation:**

When goods are not properly available in the market, customers may be forced to pay higher prices, leading to their exploitation.

## OBJECTIVES OF LOGISTICS

### 1. Rapid Response

Rapid response means delivering the required product to the customer in the **shortest possible time**. It helps the company respond quickly to customer orders and reduces the need for excessive inventory.

### 2. Minimum Variance

Variance means the **difference between actual and expected logistical performance**. It may occur due to late orders, production problems, damaged goods or incorrect delivery. Logistics aims to minimise such variance.

### 3. Minimum Inventory

Inventory involves costs such as transportation, warehousing and material handling. Therefore, logistics aims to maintain the **lowest possible inventory** while still satisfying customer requirements.

### 4. Movement Consolidation

Movement consolidation means **combining small shipments into one large shipment**. It helps reduce transportation costs as the cost per unit is generally lower for larger shipments.

### 5. Quality

Logistics aims to continuously **improve the quality of logistical activities**. Correct and damage-free delivery avoids rework, replacement and additional transportation costs.

### 6. Life Cycle Support

A company must support its product even after it is sold. This includes **after-sales services, warranty, repairs and maintenance**. It also includes reverse logistics, where products are moved from customers back to the company for repair, replacement, recycling or disposal.

### 7. Minimum Product Damage

Logistics aims to ensure that products reach customers **safely without damage**. Proper packing and packaging protect products during transportation and storage. For example, tetra packs help prevent damage to beverages.

## LOGISTICAL FUNCTIONS / LOGISTICS MIX

Logistics includes different activities that help in **moving and storing goods and delivering them to customers at the right time and place**. The main functions are **Order Processing, Inventory Management, Warehousing, Transportation, Material Handling, Packaging, Information Management and Customer Service**.

### 1. Order Processing

Order processing starts when a customer places an order. The company checks whether the goods are available, checks the customer's credit and confirms the order. Computers can help in processing orders faster.

### 2. Inventory Management

Inventory management means **keeping the right amount of stock** to meet customer needs. The company should avoid keeping too much stock because it increases costs.

### 3. Warehousing

Warehousing means **storing goods until they are needed or sold**. A warehouse helps in keeping goods safe and making them available when required. It can be private, public or contract-based.

### 4. Transportation

Transportation means **moving goods from one place to another**. It connects the company with suppliers and customers. A good transport system should have **low cost, proper speed and regular delivery**.

### 5. Material Handling

Material handling means **moving, storing and protecting materials** from the time they are received until they reach the customer. It helps in saving time, space and effort.

### 6. Packaging

Packaging means **covering and protecting products** during storage and transportation. It also makes handling and storage easier. For example, drinks are packed in bottles or tetra packs.

### 7. Information Management

Information management means **collecting and using the right information** for logistics activities. It helps in demand forecasting, order processing and making better decisions.

### 8. Customer Service

Customer service means activities done to **satisfy customers**. It ensures that the right product reaches the right customer, at the right place and time, in the right quantity and condition.

## LOGISTICAL COMPETENCY – INTRODUCTION

The two main objectives of logistics are:

- **To minimise overall logistics cost.**
- **To provide superior customer service.**

However, providing better customer service may increase logistics costs, while reducing costs may affect customer service. Therefore, a company needs to maintain a **proper balance between cost and customer service**.

**Logistical competency** means the ability of a company to provide **better customer service at the lowest possible cost**. It helps the company use its logistics activities effectively and gain a better position in the market. The main work of logistics is to **coordinate activities like transportation, warehousing, material handling and inventory management**. This helps the organisation achieve logistical competency and improve customer service and reduce costs.

There are **7 parameters** for developing logistical competency:

#### 1. Network Design

Network design means deciding **which facilities are needed, how many are needed, where they should be located and what work they should perform**. It includes plants, warehouses, distribution centres and retail stores. Proper network design helps reduce logistics costs and improve customer service.

#### 2. Information Management

Information management means using **correct and timely information** for logistics activities. It helps in demand forecasting, order processing and making better decisions. Accurate information improves logistical performance.

### 3. Transportation

Transportation means **moving goods from one place to another**. It connects the company with its suppliers and customers. Logistical competency can be improved by choosing the proper mode of transport, number of vehicles, delivery routes, etc.

### 4. Inventory Management

Inventory management means keeping the **right level of stock** to meet customer requirements while keeping costs low. It includes decisions about safety stock, lead time and replenishment of stock. Proper forecasting and stock control improve logistical competency.

### 5. Warehousing

Warehousing means **storing goods until they are needed or sold**. Logistical competency can be improved by deciding the number, location and type of warehouses and providing suitable services.

### 6. Material Handling

Material handling means **moving, storing and protecting materials** at the right place and time. Proper equipment, proper scheduling and regular maintenance help improve material handling.

### 7. Packaging

Packaging helps **protect goods and makes their handling and transportation easier**. Consumer packaging focuses on convenience and product appeal, while logistical packaging focuses on handling and protection. Choosing the proper type of packaging helps achieve logistical competency.

## REVERSE LOGISTICS

### Meaning of Reverse Logistics

Reverse logistics means **moving goods from the customer back to the company**. In normal logistics, goods move from the company to the customer, while in reverse logistics, the flow is reversed.

Reverse logistics has become important due to **environmental laws, increasing consumer awareness and strict quality standards**. Companies also use good reverse logistics services to improve customer satisfaction and remain competitive.

## SCOPE OF REVERSE LOGISTICS

### 1. Return of Unsold Goods

Unsold goods may be returned to the company when they are not sold. For example, **newspapers and magazines** can be returned.

### 2. Refilling / Reusable Packaging

Some products use packaging that can be used again. The packaging is returned to the company for refilling. For example, **milk bottles, soft drink bottles and gas cylinders**.

### 3. Repairs and Refurbishing

Products may be returned for **repair or refurbishing** due to damage, defects or poor performance. This is common with consumer durable products during the warranty period.

### 4. Product Recall

Product recall means **calling back products already sold to customers** because of manufacturing defects, safety issues, expiry or environmental hazards.

## 5. Recycling

Used products and materials such as **newspapers, cans and packaging waste** may be returned for recycling. Recycling helps reduce environmental pollution.

## 6. Scientific Disposal of Waste

Some products need to be returned for **safe and proper disposal**. Strict laws in many countries require companies to dispose of waste scientifically. For example, electronic goods like laptops need proper disposal.

## GREEN LOGISTICS

### Meaning of Green Logistics

Green logistics means **reducing the harmful effects of logistics activities on the environment and society**. It includes both forward and reverse movement of goods. Its main aim is to maintain a **balance between economic growth and environmental protection**.

### How to Achieve Green Logistics

#### 1. Network Optimization:

Companies should plan transportation routes and combine loads properly. This reduces the number of trips, fuel consumption and pollution.

#### 2. Alternative Mode of Transportation:

Companies should use **railways and other cleaner modes of transport** instead of depending mainly on road transport. This helps reduce emissions.

#### 3. Alternative Fuel:

Companies can use cleaner fuels and vehicles such as **CNG, bio-fuels, battery-operated and electric vehicles** to reduce pollution.

#### 4. Building Bypass Roads:

Bypass roads help vehicles avoid crowded areas and maintain better speed. This reduces fuel consumption and harmful emissions.

#### 5. Reduction in Paper Usage:

Companies can reduce the use of paper by using **digital bills, receipts and documents**. This helps save resources and reduce waste.

#### 6. New Fleet Induction:

Old vehicles should be replaced with **newer, fuel-efficient and low-emission vehicles**. This helps reduce fuel consumption and pollution.

#### 7. Waste Recycling and Scientific Disposal:

Used products, packaging and other waste should be **recycled and disposed of properly**. This reduces environmental pollution and ensures safe disposal.

### Examples of Green Logistics

**Kodak** designs its camera packaging so that materials can be easily recovered and recycled.

**FedEx** uses **hybrid delivery vehicles** to reduce fuel consumption and emissions.

## OBJECTIVES OF SUPPLY CHAIN MANAGEMENT

### 1. Maximise Overall Profitability

The main objective of SCM is to **increase the overall profit** of the supply chain by reducing costs and providing products efficiently to customers.

### 2. Enhance Customer Service

SCM aims to provide **better and efficient service** to customers. It helps avoid service failures and increases customer satisfaction.

### 3. Reduce Inventory Cost

Better coordination among supply chain members helps maintain the **right level of inventory**. This reduces inventory carrying costs.

### 4. Reduce Warehousing Cost

When inventory is reduced, **less storage space is required**. This helps reduce warehousing costs.

### 5. Reduce Transportation Cost

Proper planning and efficient transportation help **reduce transportation costs**. Economies of scale and economies of distance can also be used.

### 6. Reduce Lead Time

SCM aims to reduce the **time taken from receiving an order to delivering the product**. Good communication and coordination help reduce lead time.

### 7. Minimise Variance

Standardisation of activities and processes helps reduce the **difference between actual and expected results**.

### 8. Ensure On-time Delivery

All members of the supply chain should work together effectively to ensure that **products reach customers on time**.

### 9. Reduce Working Capital

Efficient SCM reduces inventory, warehousing and transportation costs. This helps the business **reduce the amount of working capital required**.

## MEANING OF CUSTOMER SERVICE

Customer service is a **series of activities performed to meet customer requirements and increase customer satisfaction**. It starts with receiving the customer's order and ends with delivering the product to the customer. In simple words, customer service means **providing the right product, at the right time and in the right condition** to satisfy the customer. Good customer service helps a business maintain its reputation and avoid loss of customers. Customer satisfaction is closely related to **good logistical performance**, because even a good-quality product will not satisfy the customer if it does not reach them properly and on time.

## OBJECTIVES OF CUSTOMER SERVICE

### 1. Ensuring Perfect Order

Perfect order means completing the **entire order correctly the first time**. The aim is to achieve maximum customer satisfaction without mistakes.

### 2. Improve Customer Satisfaction

Good customer service provides accurate and effective service according to customer needs. This helps the customer feel satisfied with the product and service.

### 3. Increase Sales

Satisfied customers are more likely to buy again and recommend the company to others. This helps the company **increase its sales**.

### 4. Retain Existing Customers

Good customer service keeps customers satisfied and prevents them from **switching to competitors**. Thus, it helps retain existing customers.

### 5. Improve Market Position

Good customer service creates **customer satisfaction, loyalty and confidence**. This helps the company increase sales and improve its market position.

### 6. Retain Customer Loyalty

Customers tend to continue doing business with companies that provide good service. Therefore, efficient customer service helps the company **build and retain customer loyalty**.

## ELEMENTS OF CUSTOMER SERVICE

Customer service includes activities performed by an organisation to **make the customer's buying experience better**. Good customer service increases customer satisfaction, builds loyalty and helps increase sales.

The elements of customer service are divided into **three phases**:

### I. PRE-TRANSACTION ELEMENTS

These are activities performed **before the customer places an order**.

#### 1. Customer Service Policy

The company should have a **clear customer service policy** which guides employees and informs customers about the services provided by the company.

#### 2. System Flexibility

The organisation should be able to **adapt to changes in customer needs, tastes and business conditions**.

#### 3. Organisation Structure

The organisation should have a **proper structure** for smooth communication of policies, performance and corrective actions.

### II. DURING-TRANSACTION ELEMENTS

These are activities performed **when the customer is placing or receiving an order**.

### 1. Product Availability

The required product should be **available when the customer needs it**. Timely availability increases customer satisfaction.

### 2. Correct Order Status Information

The company should provide **correct and timely information** about the status of the customer's order

### 3. Consistency in Order Cycle Time

Order cycle time is the **time between placing an order and receiving it**. This time should be short and consistent.

### 4. Substitute Products

If the main product is not available, the company should **offer a suitable substitute product** to the customer.

## III. POST-TRANSACTION ELEMENTS

These are activities performed **after the customer has purchased the product**.

### 1. After-Sales Service

It includes **installation, warranty, guarantee, repair and maintenance** services after the sale.

### 2. Product Tracing

The company should be able to **trace and recall products** if there are quality or safety problems.

### 3. Customer Grievance Redressal

The company should **listen to customer complaints and take proper action** to solve them.

### 4. Product Loaning

The company may provide a **temporary replacement product** when the original product is being repaired or serviced.

## LEVELS OF CUSTOMER SERVICE

### 1. Customer Service as an Activity

Customer service is treated as an **activity performed to satisfy customer needs**, such as order processing and invoicing.

### 2. Customer Service as a Performance Measure

Customer service is treated as a **measure of logistics performance**, such as the number of perfect orders delivered.

### 3. Customer Service as a Philosophy

Customer service is treated as a **commitment of the company**. The company makes customer service an important part of its policies and overall performance.